FASHION PRODUCTS IDENTIFICATION USING ARTIFICIAL NEURAL NETWORK (ANN)

Automatically Detect and Classify Fashion Products from Images

PROJECT AIM

To develop an AI based system that can automatically identify and classify fashion products using Artificial Neural Network (ANN) and the Fashion-MNIST dataset.

PROBLEM STATEMENT

In clothing stores and warehouses, products are manually sorted into categories which is time consuming, error prone and labour intensive. An automated system is required that can recognize and classify different clothing items accurately and efficiently.

DATASET DETAILS

70,000	28×28	10	60,000	10,000
Total Images	Image Size	Categories	Training Images	Testing Images
Grayscale Images	Pixels	Clothing Items	(85.7%)	(14.3%)

FASHION-MNIST OVERVIEW

Fashion-MNIST is a dataset of Zalando's article images consisting of 10 different categories of clothing products. Each image is grayscale, fixed-size (28×28 pixels) and represents a single product.

0	1	2	3	4	5	6	7	8	9
T-shirt/Top	Trouser	Pullover	Dress	Coat	Sandal	Shirt	Sneaker	Bag	Ankle Boot

SYSTEM WORKFLOW

1 DATASET

Fashion-MNIST Dataset
70,000 Images
10 Categories

2 PREPROCESSING

- Resize (28×28)
- Grayscale Conversion
- Normalization (0-1)
- One-Hot Encoding

3 ANN MODEL

- Input Layer
- Hidden Layers
- Output Layer (Softmax)
- Train the Model

4 PREDICTION

Model predicts the category of the input image

5 RESULT

Identified Clothing Category with High Accuracy

KEY BENEFITS

- Saves Time
- Reduces Human Errors
- Improves Efficiency
- Supports Smart Retail & Warehouse Automation

FASHION PRODUCTS IDENTIFICATION
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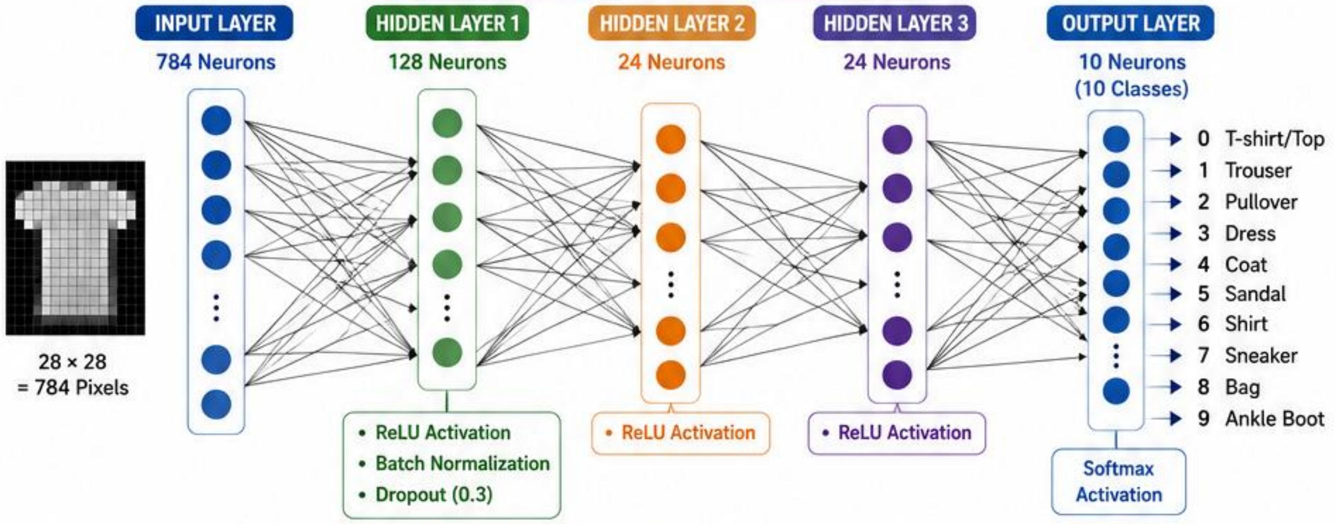
PAGE 2 : ANN MODEL ARCHITECTURE & TRAINING PROCESS

ANN MODEL ARCHITECTURE



Input Size	Hidden Layers	Activation Function	Output Function	Total Classes
784	3	ReLU	Softmax	10
(28×28)	(128, 24, 24)	(Hidden Layers)	(Multi-class)	(Clothing Items)

DATA PREPROCESSING PIPELINE

1. **RESIZE**
   Resize all images to 28 × 28 pixels
2. **GRAYSCALE CONVERSION**
   Convert RGB images to grayscale
3. **NORMALIZATION**
   Normalize pixel values between 0 and 1 (pixel / 255.0)
4. **FLATTEN (RESHAPE)**
   Convert 28×28 image into 1D array of size 784
5. **ONE-HOT ENCODING**
   [0 0 1 ... 0]
   Convert labels into one-hot encoded vectors (10 classes)

READY FOR MODEL TRAINING

MODEL TRAINING PROCESS

Training Data
60,000 Images
(85.7%)

Model Compilation
Optimizer : Adam
Loss : Categorical Crossentropy
Metrics : Accuracy

Training
Epochs : 30
Batch Size : 128

Validation
Validation Split : 0.15
(10,000 Images for Testing)

Save Model
Best performing model is saved for prediction

MODEL BUILDING (Keras)

```
model = Sequential()
model.add(Dense(128, input_dim=784))
model.add(Activation('relu'))
model.add(BatchNormalization())
model.add(Dropout(0.3))
model.add(Dense(24))
model.add(Activation('relu'))
model.add(Dense(24))
model.add(Activation('relu'))
model.add(Dense(10))
model.add(Activation('softmax'))

model.compile(optimizer='adam',
     loss='categorical_crossentropy',
     metrics=['accuracy'])
```

Why This Architecture?

- ReLU helps in faster convergence
- Batch Normalization improves stability
- Dropout reduces overfitting
- Softmax gives probability for each class

DATA FLOW IN ANN MODEL

INPUT IMAGE
(28×28, Grayscale)

→ **PREPROCESSING**
Resize, Grayscale, Normalization, Reshape, One-Hot

→ **FEATURE LEARNING**
ANN learns patterns from the training images

→ **CLASSIFICATION**
Model classifies the image into one of the 10 classes

→ **OUTPUT**
Predicted Class with Probability (Softmax)

KEY POINTS

- ANN learns complex patterns from pixel values.
- Model predicts the category of unknown fashion products.
- High accuracy achieved with proper preprocessing and training.

TECHNOLOGIES USED
Python | TensorFlow / Keras | NumPy | Pandas | Matplotlib | Google Colab